REVIEW PROBLEMS

Please note that this is NOT a practice exam – the length and style of questions are not necessarily indicative of the length and style of the actual exam. This is just a review of problems addressing some of the concepts you learned so far.

QUESTION 1

Tennis players often spin a tennis racquet and observe whether it lands with the logo facing up or down, to determine who serves first. But is this really a 50-50 process, equally likely to land with the logo facing up or down? To investigate this, a tennis player spun his racquet 100 times, and he obtained 46 "up" (success) and 54 "down" (failure) results. The research question being investigated is whether the data provide much evidence against believing that spinning the racquet is really a fair 50-50 process.

a. The 95% confidence interval for the parameter of interest is (0.322, 0.598). Interpret this 95% confidence interval.

b. How do you expect the confidence interval to change if the sample size was 200 and the sample proportion was still 0.46?

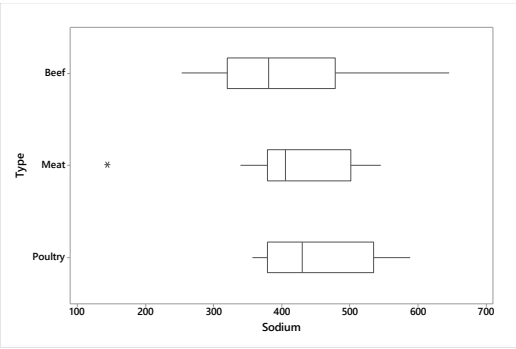
QUESTION 2

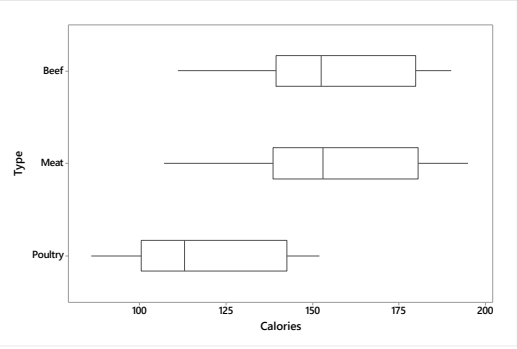
Suppose that Samira selects a random sample of 500 residents of San Luis Obispo and asks whether or not the person has used Zoom in the past week. Luis selects a random sample of 500 residents of Los Angeles (which has a much larger population than San Luis Obispo) and asks the same question. Assume sample statistic stays the same. Whose survey will have a smaller margin-of-error?

A. Samira
B. Luis
C. They will have the same margin-of-error

QUESTION 3

The following graphs display data from a sample of hot dogs of three different types:





a) Do the three types of hot dogs differ more substantially with respect to calories or sodium? Choose one.

A) Calories
B) Sodium

b) Which type of hot dog is most different from the other two types? Choose one.

A) Beef
B) Meat
C) Poultry

c) Estimate the median value of calories among poultry hot dogs.

d) Estimate the inter-quartile range of calories among poultry hot dogs.

e) Estimate the largest value of sodium contents among all of the hot dogs.

QUESTION 3

The 2004 General Social Survey (GSS) interviewed a random sample of adult Americans. For one question the interviewer asked: "From time to time, most people discuss important matters with other people. Looking back over the last six months–who are the people with whom you discussed matters important to you? Just tell me their first names or initials." The interviewer then recorded how many names or initials the respondent mentioned. Results are tallied in the following table, and the mean was 1.987 friends and the standard deviation was 1.7708 friends.

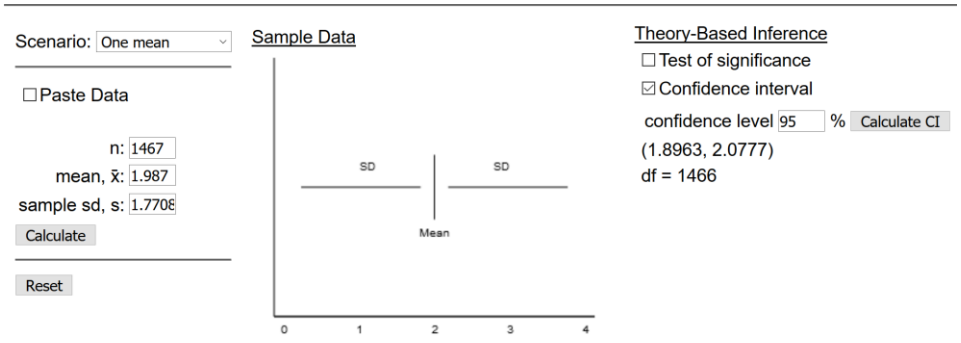
Number of Close Friends	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Count (Number of Respondents)	397	281	263	232	128	96	70	1467

a. Identify the observational unit.

b. Identify the variable in the study. Is the variable categorical or quantitative?

The screenshot below shows the computation of the 95% confidence interval using the Theory-based Inference applet.

Theory-Based Inference



c. Which of the following is the best interpretation of this confidence interval and its confidence level?
   a. Ninety five percent of all adult Americans in this sample reported a number of close friends within this interval.
   b. If you took another sample of 1467 adult Americans, there is a 95% chance that its sample mean would fall within this interval.

c. You can be 95% confident that the mean number of close friends in the population of adult Americans is between the endpoints of this interval.
d. This interval captures the number of close friends for 95% of the people in the population of adult Americans.

d. Look at the previous part and explain why choice B is incorrect.

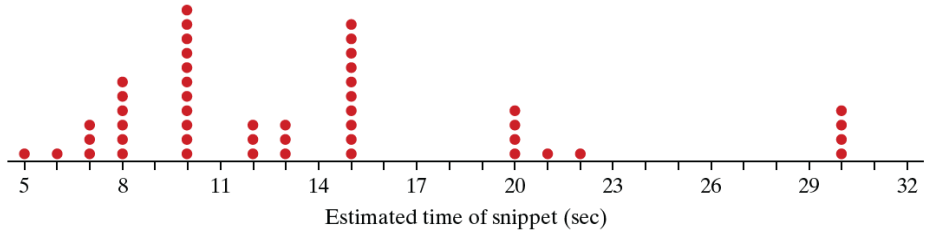
QUESTION 4

Does it ever seem like time drags on (perhaps during one of your least favorite classes) or time flies by (like a summer vacation)? Researchers asked 48 college students at a university in the US to be subjects in the experiment. The participants were asked to listen to a 10-second snippet of a song, and after it was over, they were asked to estimate how long the song snippet lasted. Afterward, they simply asked the subjects how long they thought the song snippet lasted (The subjects did not know in advance that they would be asked this question). They wanted to see whether students could accurately estimate the length of this short song segment.

a) What is the research question of this study?

b) State null and alternative hypotheses for this study using symbols.

Here is the dotplot of this study showing the results (in seconds) for 48 students trying to estimate the length of a 10-second song snippet. The sample mean is 13.71 seconds, the median is 10 seconds, and the sample standard deviation is 6.50 seconds.



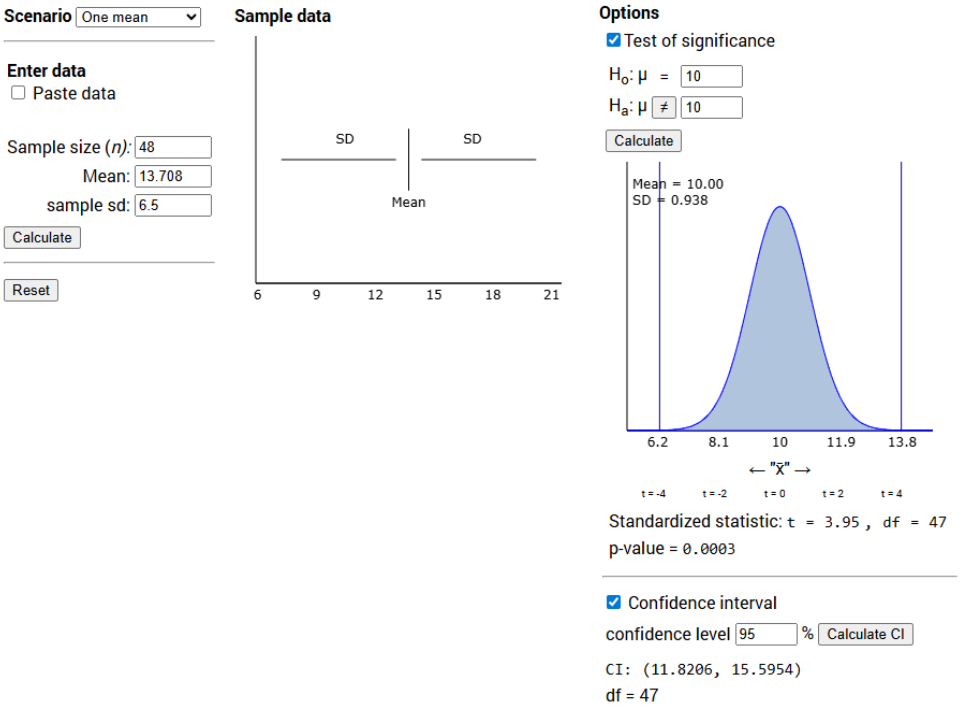
c) Describe this distribution using the context of the study.

d) Is it possible that this overestimate (sample mean 13.71 seconds larger than 10 seconds) just come from random chance if, on average, the population is accurate when estimating the length of a short song snippet ($\mu = 10$)?

e) Are the validity conditions met to conduct a one-sample t-test? Explain.

f) Regardless of your answer to the previous question, calculate the standardized statistic. Show your work.

g) Suppose the researcher conducted the analysis below. Is the analysis correct? Explain.



h) Interpret the p-value.

i) What is the confidence interval in the screenshot above estimating?
   A) The sample mean
   B) The population mean
   C) The difference in sample means
   D) The difference in population means
   E) None of the above.

j) To which population can you generalize the results of this study?

k) State your conclusion in context. Make sure to refer to the parameter and the population.

QUESTION 5

A study examined whether a nicotine lozenge can help a smoker to quit. The research article reports on many background variables, such as age, weight, gender, number of cigarettes smoked, and whether the person made a previous attempt to quit smoking (Shiffman et al., 2002). Suppose the researchers want to compare the distributions of the **background variables** between the two treatment groups (nicotine lozenge or placebo lozenge).

a. For each of the five variables listed, indicate whether it calls for a comparison of means or a comparison of proportions.

b. In the context of making sure the groups are comparable, would the researchers hope to reject the null hypotheses or fail to reject the null hypotheses in these tests? Explain.

QUESTION 6

In a study published in the July 2003 issue of the journal Psychosomatic Medicine, researchers reported that people who tend to think positive thoughts catch a cold less often than those who tend to think negative thoughts (Cohen et al., 2003). The scientists recruited over 300 initially healthy volunteers, and they first interviewed them over 2 weeks to gauge their emotional state, eventually assigning them a score for their emotions. Subjects are then classified as having positive or negative emotions. Then the researchers injected rhinovirus, the germ that causes colds, into each subject's nose. The subjects were then monitored for the development of cold-like symptoms. Subjects having negative emotions were three times more likely to catch a cold than those having positive emotions.

a. Identify the explanatory and response variables in this study. Classify each as categorical (also binary?) or quantitative.

b. Is this an observational study or an experiment? Explain. [Hint: Ask yourself whether or not the explanatory variable was assigned by the researchers.]

c. Based on this study and the substantial differences in likelihood of catching a cold between these groups, can you draw a cause-and-effect conclusion between emotional outlook and likelihood of catching a cold? If yes, explain why. If not, identify a potential confounding variable and explain how it is confounding in this study.

QUESTION 7

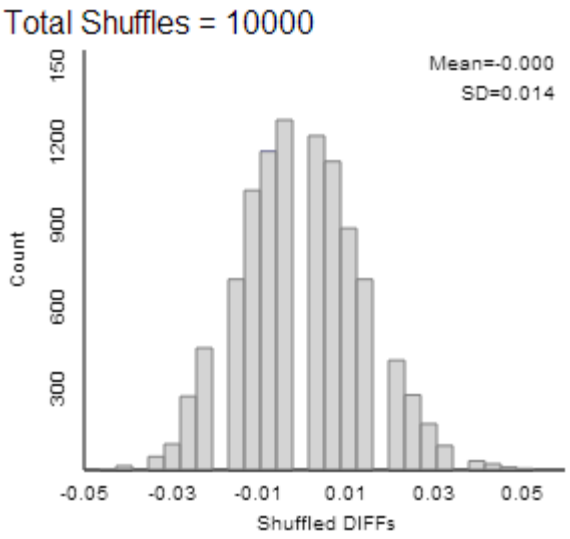
For several years in the 1990s, Kristen Gilbert worked as a nurse in the intensive care unit (ICU) of the Veteran's Administration hospital in Northampton, Massachusetts. Over the course of her time there, other nurses came to suspect that she was killing patients by injecting them with the heart stimulant epinephrine. Gilbert was eventually arrested and charged with these murders. Part of the evidence presented against Gilbert at her murder trial was a statistical analysis of 1641 eight-hour shifts during the time Gilbert worked in the ICU investigating whether there was a higher risk of death when gilbert was working. For each of these shifts, researchers recorded two variables: whether or not Gilbert worked on the shift, and whether or not at least one patient died during the shift.

	Gilbert	**No Gilbert**	**Total**
At least one death	40	34	74
No deaths	217	1350	1567
Total	257	1384	1641

a. Identify the observational units in this study.

b. Which variable is the explanatory variable?

c. The graph below simulates randomly assigning the 1641 shifts to be with or without Gilbert. Based on the following output, will you reject or fail to reject the null hypothesis?



d. Provide a conclusion in context, make sure you refer to the parameter and population.

e. Is this convincing evidence that Gilbert *caused* a higher probability of at least one death during shifts when she worked? Explain.

f. Suggest another way of measuring how extreme the observed statistic is in the null distribution. Calculate and show your work.

QUESTION 8

A researcher studied whether AZT helps to reduce transmission of AIDS from mother to baby. Of the 180 babies whose mothers had been randomly assigned to receive AZT, 13 babies were HIV-infected, compared to 40 of the 183 babies in the placebo group. Consider HIV-infected as "success". For any calculations use the order AZT – PLACEBO.

a. Create a table with the results from the study. Display the explanatory variable as columns and the response variable as rows.

b. Create a segmented bar graph to display these results. Comment on what the graph reveals.

c. Are the validity conditions met to conduct a two-sample z-test?

d. Does this study involve random sampling from populations or random assignment to treatment groups or both?

e. Conduct an appropriate test of significance to determine whether the data provide convincing evidence that AZT is more effective than a placebo for reducing mother-to-infant transmission of AIDS. Report the hypotheses, statistic, and p-value.

f. Based on the results of your study, which error could you have committed? Type I or Type II error?

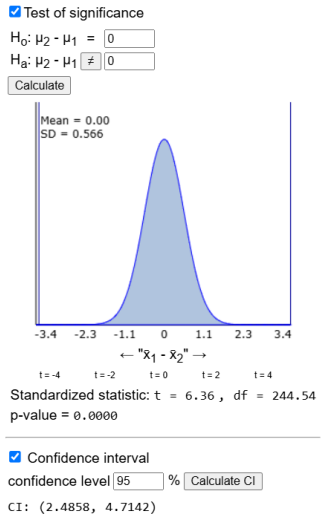
g. The 99% confidence interval for the parameter of interest is (-0.2394, -0.0533). Interpret this interval in the context of the study.

h. Summarize the conclusion that you could draw from this study (significance, causation, and generalizability).

QUESTION 9

A study was conducted to see whether the average number of friends reported differs by those with ages 18–29 and those with 65+. Assume the order: (age 65plus) – (age 18-29)

a. Interpret the output below:



b. What type of analysis was conducted in the screenshot above?
   a) Simulation-based approach
   b) Theory-based approach

c. Describe the effect (decrease, increase, or remain the same) on the p-value of the significance test, and explain your reasoning in each case, if everything remains the same except
   1. Every person 65+ and every person with age between 18-29 sampled have one more close friend than they originally reported.
   2. Both sample standard deviations are larger.
   3. Both sample sizes are larger.

QUESTION 10

The Palmer Penguins data set contains measurements taken from penguins in the Palmer Archipelago, Antarctica, from 2007 to 2009. Consider 146 Adélie penguins for which sex and body mass were recorded. Researchers want to investigate whether the population mean body mass differs for male and female Adélie penguins. For this review problem, assume that these penguins are representative of Adélie penguins in the Palmer Archipelago. Order used: Male – Female. The results of the study are summarized below.

Female: n = 73, mean = 3368.84 g, SD = 269.38 g
Male: n = 73, mean = 4043.49 g, SD = 346.81 g

a. Identify one observational unit.

b. Identify the explanatory variable, and response variable.

c. Calculate the statistic. Use appropriate symbols to represent the statistic.

d. State the hypotheses in symbols.

e. Interpret any symbols you used in the previous question.

f. Suppose the standardized statistic to determine whether the data provide convincing evidence that the population means differ for male and female Adélie penguins is 13.13. State the conclusion in context. Make sure you refer to the parameter and population.

g. A 95% confidence interval for the parameter of interest is (573.0, 776.3). Interpret this interval in context.

h. Can you make a causal claim based on the results of this study?